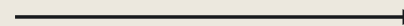


# SLEEP-TIME COMPUTE

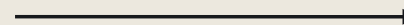
## 기억 생애주기의 다음 인프라

wake 에서 포착하고 · sleep 에서 변환하고 ·  
장기기억에서 검증 · 검색 · 삭제한다

WAKE



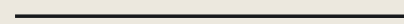
SNAPSHOT



SLEEP



MEMORY



Evidence → candidate → verified manifest

# 핵심은 ‘자는 모델’ 이 아니라 기억의 이동 규칙이다

통합 추론

언제

배포 전 | wake / test-time | delayed sleep

무엇을

replay · distill · dream · merge · verify · forget

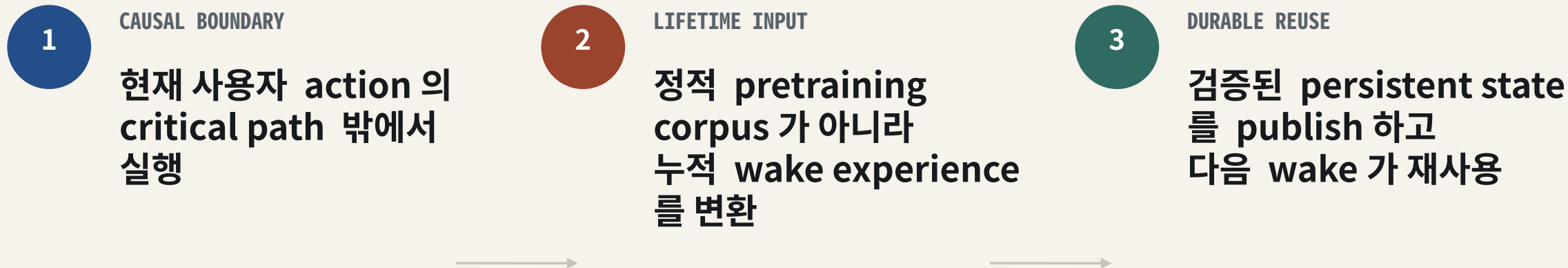
어디로

text / vector / graph → latent / adapter → shared weight

모든 이동은 provenance · capacity · rollback · deletion 을 함께  
가져야 한다

# strict sleep 은 세 조건을 동시에 만족한다

제안



따라서 online summary  $\neq$  sleep · offline corpus distill  $\neq$  deployed sleep · idle GPU  $\neq$  memory consolidation

이 정의는 scheduling label 과 computational phase 를 분리한다 .

# 앞으로의 학습은 세 phase 와 하나의 memory fabric 으로 나뉜다

통합 추론

PRE

배포 전

pretrain · post-train  
update rule 학습

WAKE

사용 중

infer · retrieve  
TTT · capture

SLEEP

휴면 경계

replay · distill  
compact · validate

MEMORY

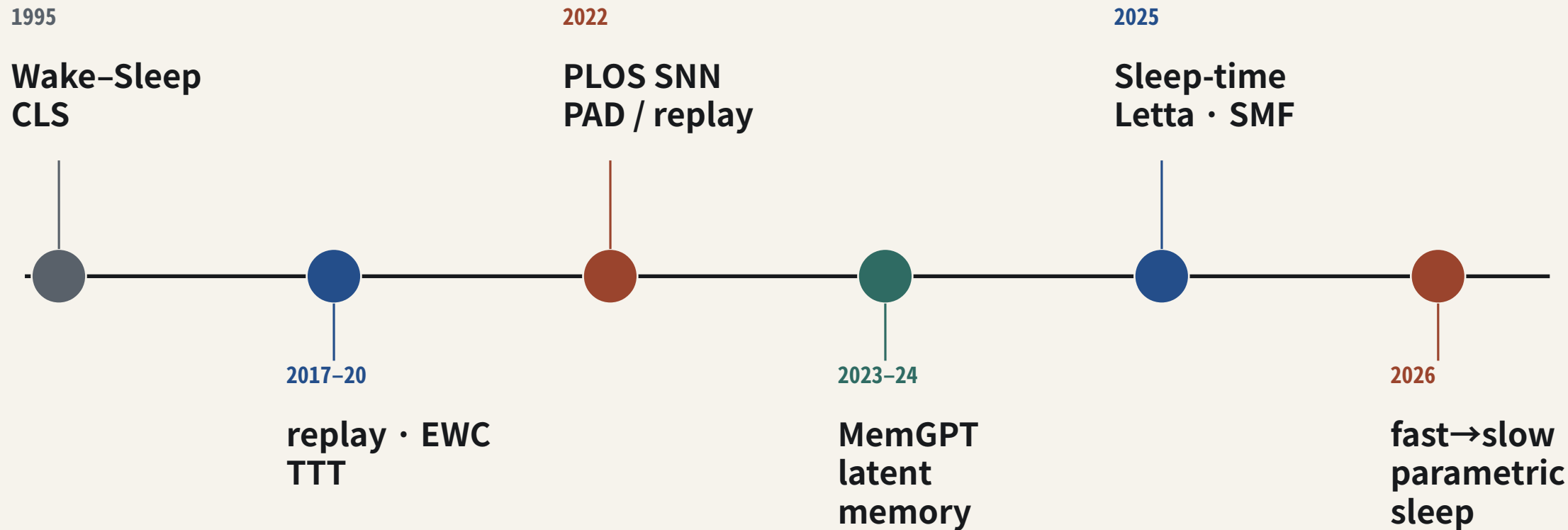
지속 상태

canonical evidence  
views · adapters

fixed base + session/user-local mutable state + shared slow state + external canonical truth

# ‘sleep-time’이라는 이름보다 계보가 먼저였다

근거



직선적 진보가  
아니다

online replay · query-time compilation · external lifecycle · parametric  
consolidation 은 timing 과 destination 이 다르다 .

# 생물학은 operator 후보를 준다—역할표를 복사하라는 뜻은 아니다

근거

## 전이 가능한 근거

- 경험 관련 activity 의 sleep replay
- cue·rhythm 조작이 다음날 recall 을 변화
- fast episodic ↔ slow structured system
- replay·downscaling·recombination 후보

## 전이하면 안 되는 비약

- hippocampus→cortex ‘파일 이동’
- NREM= 정답 replay / REM= 창의성의 확정
- 동물 결과를 LLM lifetime law 로 일반화
- sleep 이면 이미 지워진 정보도 복원

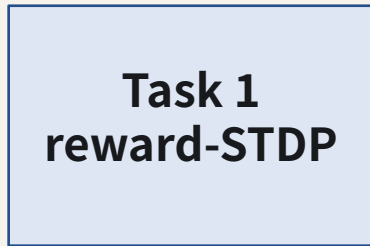
## ENGINEERING RULE

stage 이름보다 data · objective · publication · later reuse 를  
기록한다

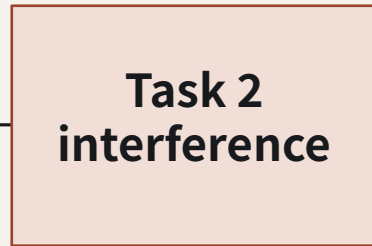
# 2022 PLOS 는 ‘학회’ 가 아니라 두 과제 SNN 저널 실험이다

근거

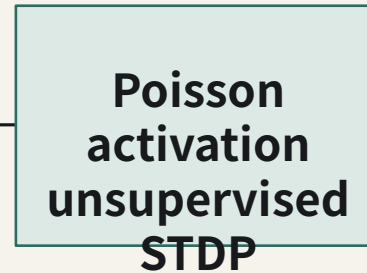
WAKE 1



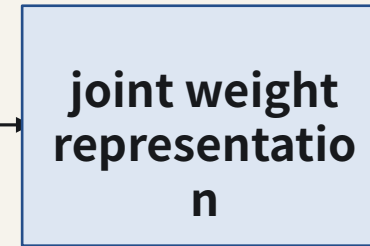
WAKE 2



SLEEP



LATER WAKE



842

neurons  
2 tasks  
≥10 inits

순차 학습

Task 1  $0.52 \pm 0.02$ 

wake↔sleep

Task 1/2  $0.70 \pm 0.03$  /  $0.68 \pm 0.05$ 

한계

old trace 가 완전히 사라진 뒤에는  
복원하지 못했다.

LLM · 다과제 · 장기용량 ·  
deletion 은 검증하지 않았다.

# PAD 는 계산 가설 , DANN 은 낮은 증거 branch 다

근거

## PAD Perturbed & Adversarial Dreaming

**NREM** episodic latent → occlusion → reconstruction

**REM** memory mix + noise → adversarial dream

범위 : CIFAR-10 / SVHN representation learning

## DANN Dream-Augmented Neural Networks

- SSRN metadata · abstract 확인
- architecture/data/seed/code 독립 감사 불가
- ‘zero forgetting’ 은 사용 금지
- 방향성 아이디어로만 유지

EVIDENCE GRADE · LOW / DIRECTIONAL

# phase 와 기억 위치는 독립 축이다

통합 추론

|           | EXTERNAL                      | LATENT / KV                      | ADAPTER           | SHARED WEIGHT |
|-----------|-------------------------------|----------------------------------|-------------------|---------------|
| WAKE      | Mem0 · Graphiti               | TTT · Titans                     | fast-LoRA         |               |
| SLEEP     | Letta sleeper<br>Auto-Dreamer | Cartridges<br>Offline Recurrence | Doc→LoRA          | Google Sleep  |
| PREDEPLOY | memory-policy<br>pretraining  |                                  | adapter generator | SMF substrate |

핵심 질문

같은 future utility 를 만들 때 어느 phase × destination 이 가장 적은 lifecycle cost 와 가장 강한 reversibility 를 가지는가 ?

# MemGPT · Letta · Mem0 · Zep 은 같은 ‘sleep’ 이 아니다

| 근거                                         |                  |                                      |               |                           |
|--------------------------------------------|------------------|--------------------------------------|---------------|---------------------------|
| 2023                                       | MemGPT           | context pressure · recursive summary | wake          | 별도 sleeper 없음             |
| 2025                                       | Letta paper      | pre-query context compilation        | sleep-time    | durable memory 와 구분       |
| 2025                                       | Letta Sleeper    | async core-memory editing            | product S     | paper benchmark 와 구분      |
| 2025                                       | Mem0 paper / OSS | ADD · UPDATE · DELETE · NOOP         | mostly wake   | mutation 과 phase 를 구분     |
| NOW                                        | Mem0 Platform V3 | asynchronous ADD-only path           | async         | later-read 경로             |
| 2025                                       | Graphiti OSS     | temporal graph invalidation          | ingest        | scheduler 미소유             |
| NOW                                        | Zep Cloud / Obs. | async later-read / observer          | async / prop. | Cloud PASS · operator 비공개 |
| paper ≠ OSS ≠ cloud ≠ proprietary operator |                  |                                      |               |                           |

# Google · Meta · Microsoft 는 서로 다른 병목을 공략한다

근거

GOOGLE / DEEPMIND

CLS · EWC ·  
Titans/Miras

fast→slow expert ·  
synthetic dream

experimental  
parametric

META / FAIR

GEM · MAS · Product-  
Key Memory

Memory Layers ·  
sparse row update

capacity & low-  
interference  
write

MICROSOFT

LongMem ·  
Generative Adapter

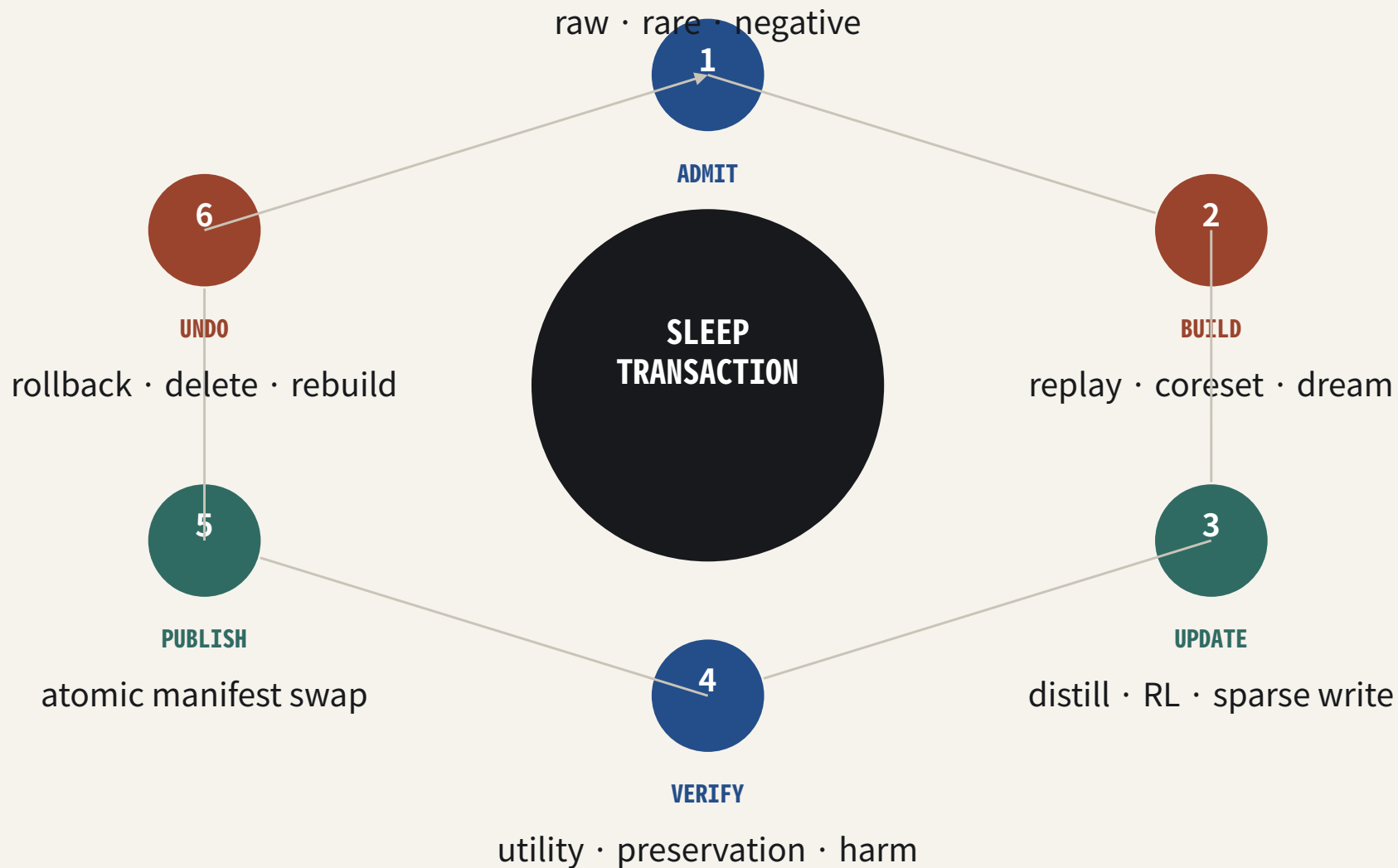
HIMA · MEMENTO ·  
RHO

external  
schedule &  
harness

비주장 · 공개된 논문 계보를 회사의 단일 제품 전략 · production deployment 로 일반화하지 않는다

# sleep 은 하나의 optimizer 가 아니라 operator pipeline 이다

통합 추론



# 좋은 dream 은 미래 query utility 를 보존한다

제안

HOLDOUT FUTURE TASK

dream likelihood 가 아니라 다음 wake utility

GROUNDING

source provenance · tool outcome · simulator

ANTI-COLLAPSE MIX

raw · teacher · student · adversarial · random

NOVEL RECOMBINATION

counterfactual · composition · edge case

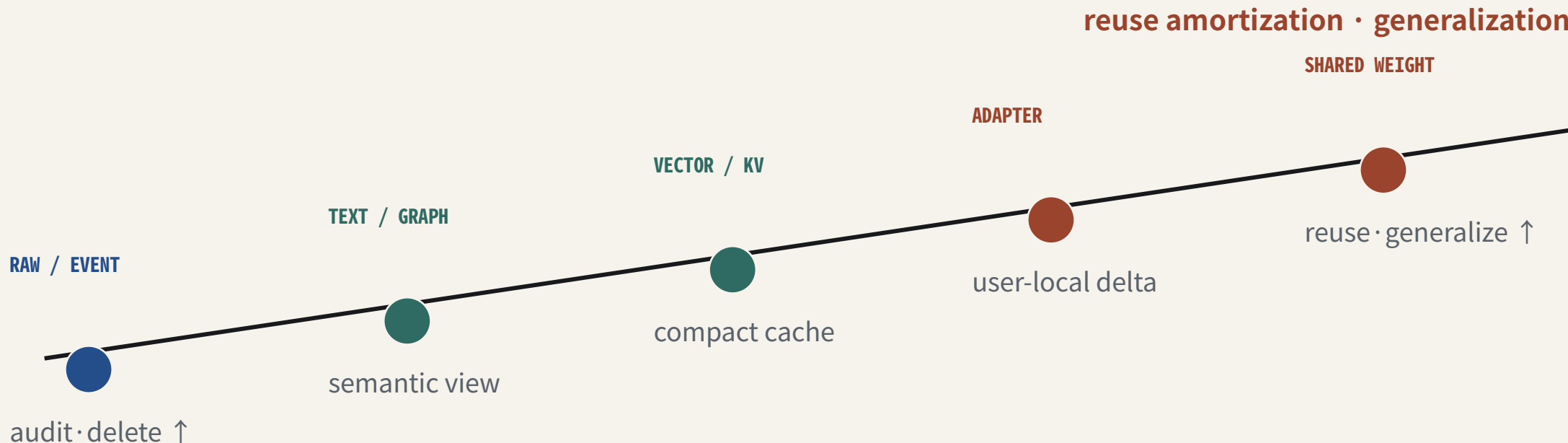
COVERAGE

old · rare · negative · correction

teacher corruption 과 omission 은 sleep 을 반복할수록 축적될 수 있다

# external-first, parameter-last 는 방어 가능한 default 다

통합 추론



reversibility · provenance ·  
correction

PROMOTE only if expected future value > compile + serve + interfere + govern + risk

# 기억용량은 최소 다섯 개이고 먼저 차는 것이 다르다

통합 추론

01

STORAGE

bytes · rows · edges · slots

02

RETENTION

old trace above recall threshold

03

ADDRESSABILITY

find and apply under live budget

04

PLASTICITY

learn new as well as fresh control

05

GOVERNANCE

provenance · delete · rollback

first binding pressure

# 근거가 사라지면 sleep 도 특정 기억을 부활시킬 수 없다

통합 추론 ———



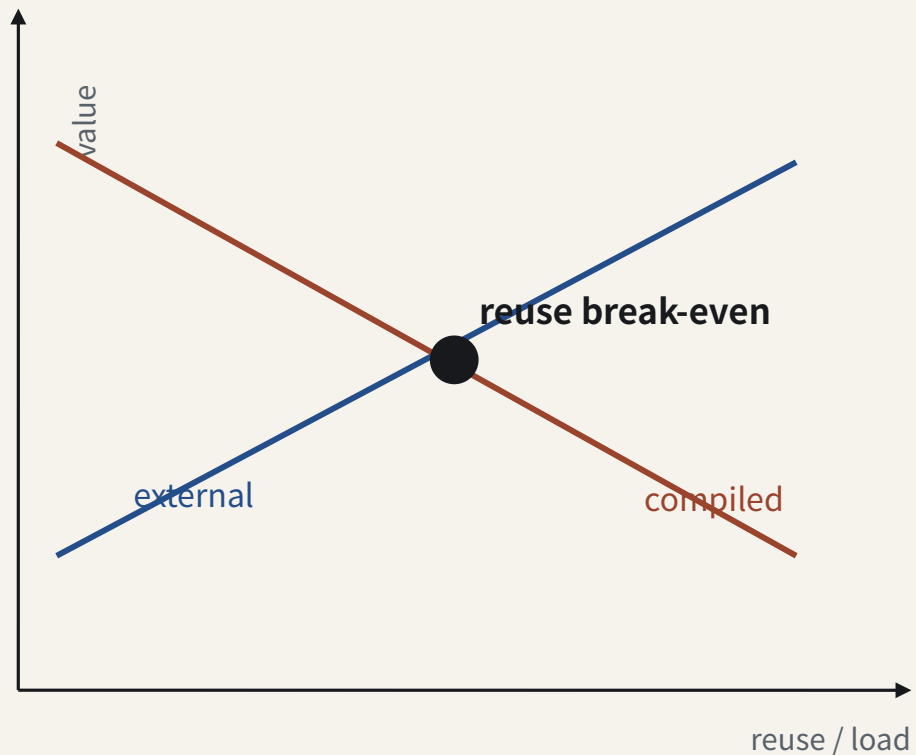
$I(X; \Omega_t) = 0 \longrightarrow \Rightarrow$  특정  $X$  의  
복원 불가

정보가 남는  
곳

raw event · replay buffer · teacher · generator · residual weight ·  
adapter · checkpoint · archive · implicit KV  
hidden carrier 도 capacity · privacy · delete ledger 에 포함한다

# 단일 $N^\alpha$ 보다 crossover atlas 가 더 가능성 높다

가설



CAPACITY KNEE

association load ×  
state budget

CADENCE

staleness × run cost  
× burst

QUEUE

arrival × service ×  
clocks

ROUTING

module isolation ×  
catalog tax

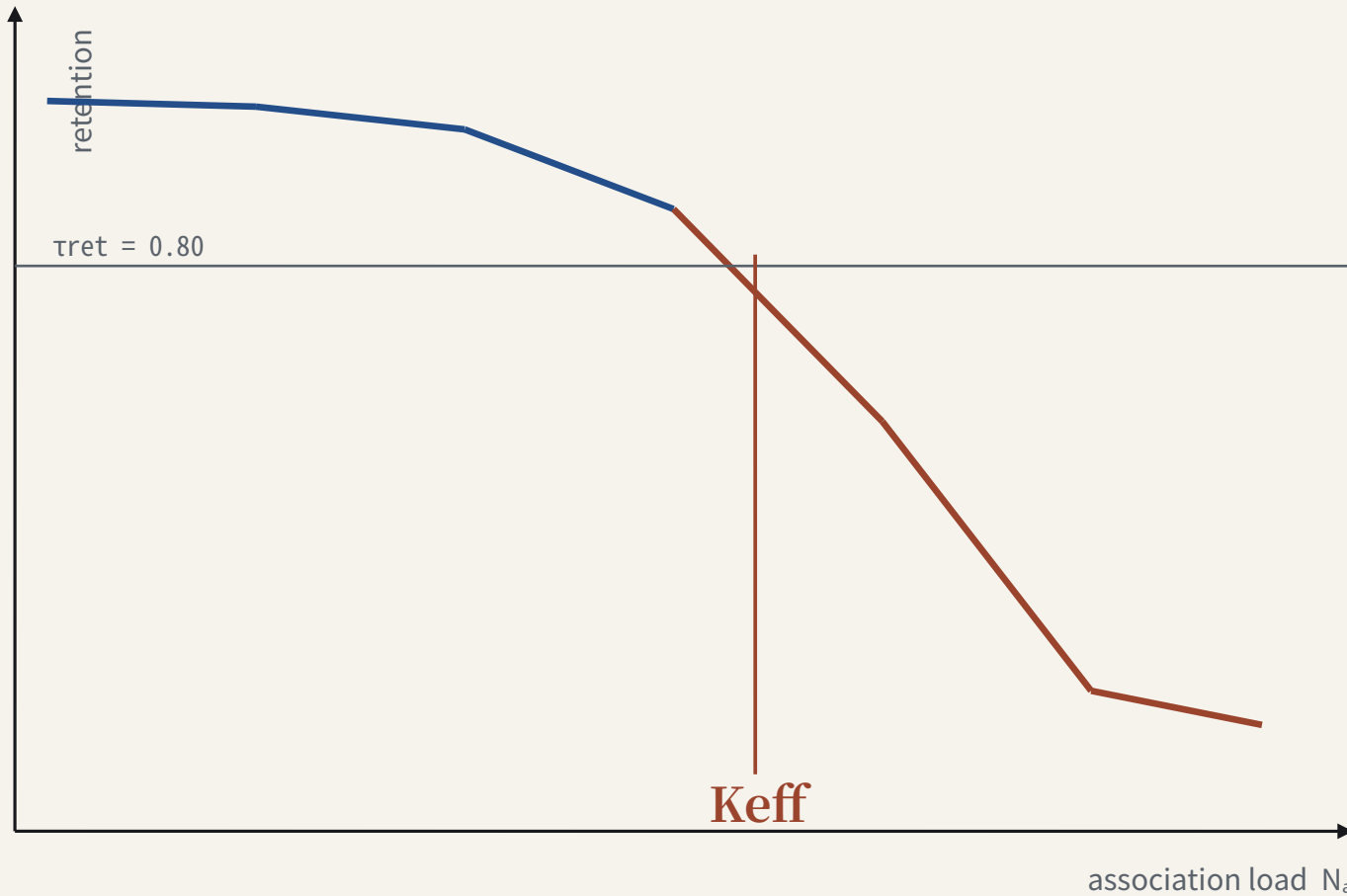
RECOVERY

compression ×  
reversibility reserve

SCHEMATIC · NOT A RESULT

# capacity 는 ‘최대 몇 개’ 보다 품질이 꺾이는 경계로 측정한다

가설



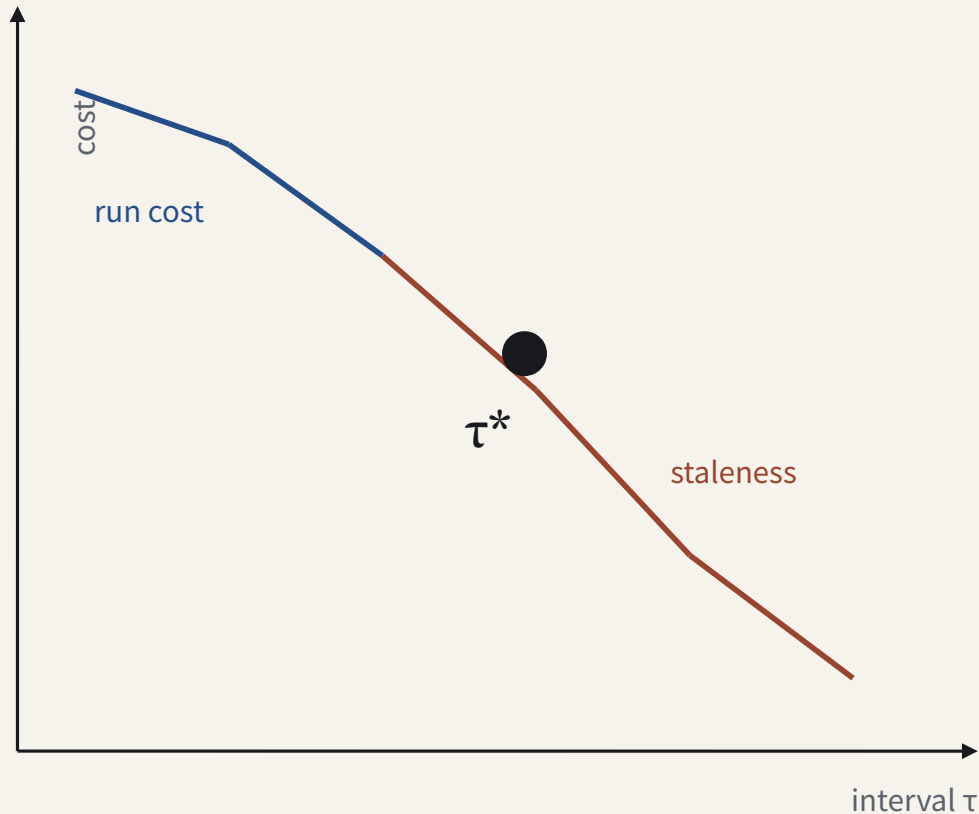
## KNEE AUTHORIZATION

- low/high bracket
- monotonic
- single crossing
- residual envelope
- shape sensitivity

else KNEE\_UNIDENTIFIED

# sleep clock 은 정확도보다 queue 와 staleness 의 공동 문제다

가설



## QUEUE CONDITION

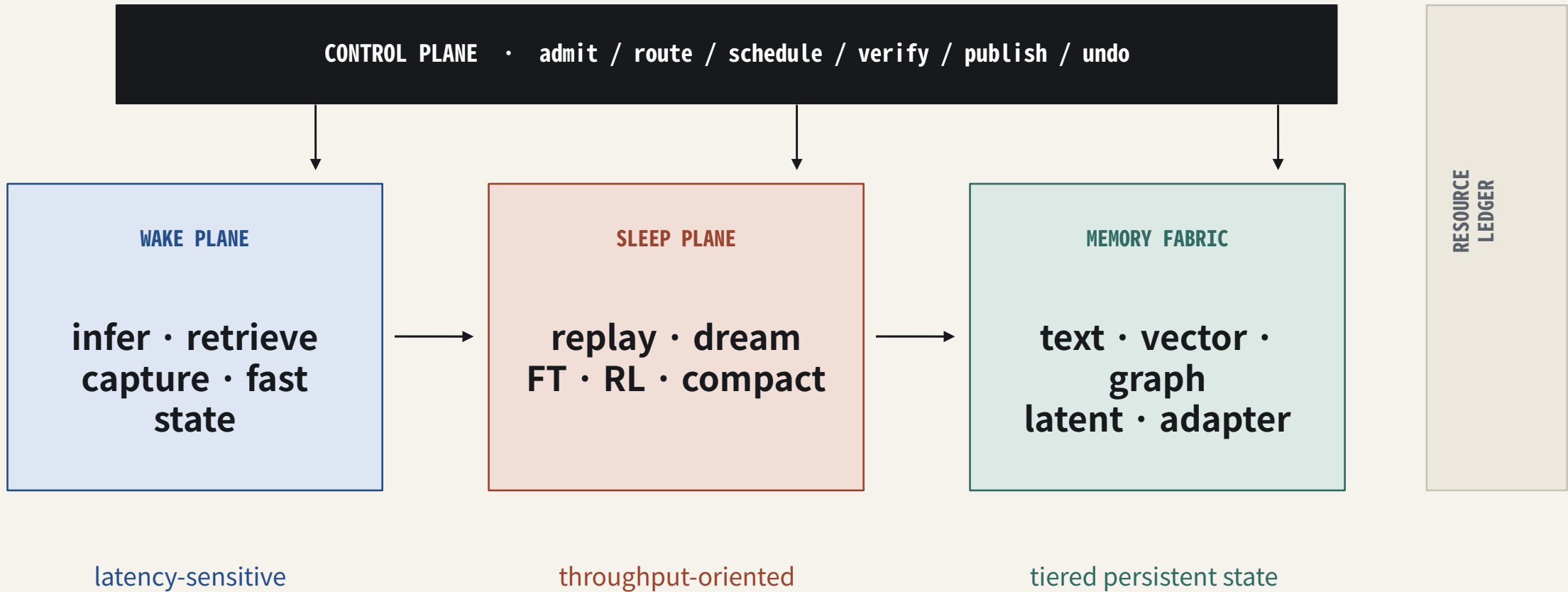
$$\rho_s = \lambda_s E[S_s] / m < 1$$

- mean-load 필요조건이지 tail 보장 아님
- GPU · IO · RAM · HBM 을 typed vector 로
- delete deadline 과 p99 job age 포함
- clock 수가 늘면 catalog · routing tax 증가

HYPOTHESIS · queue-before-storage

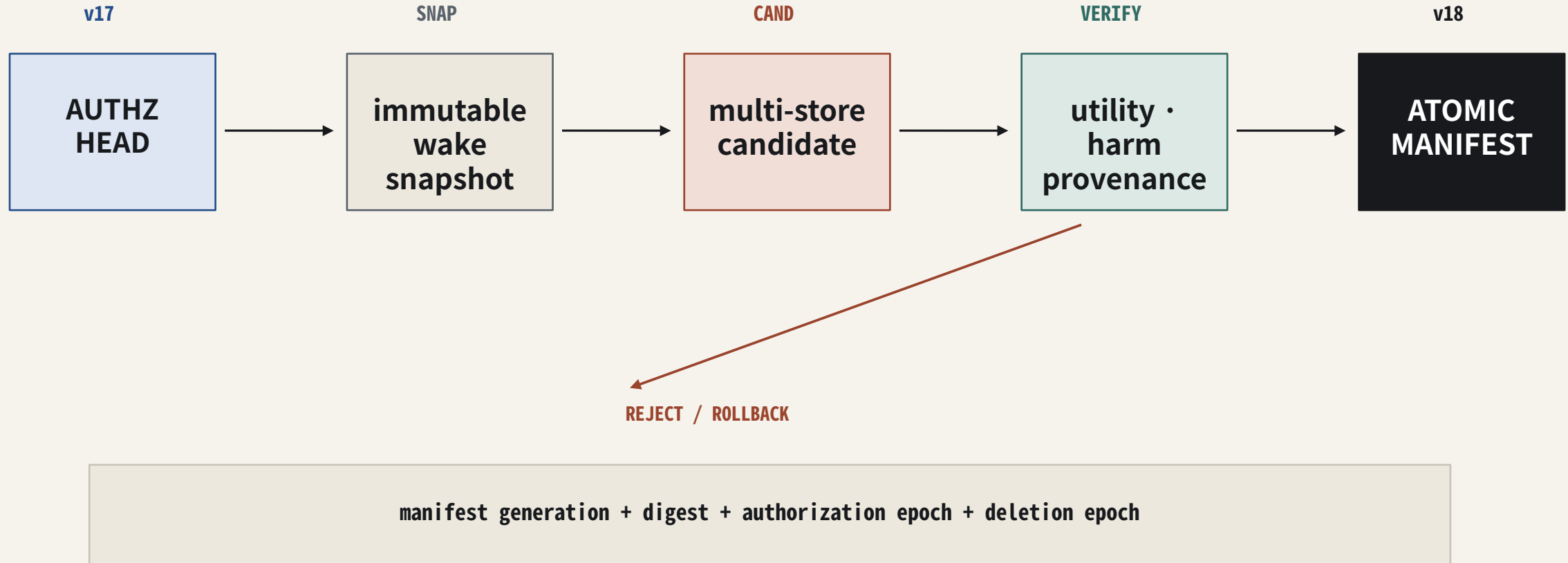
# 세 logical plane 과 하나의 control plane 이 필요하다

제안



# sleep 완료는 job 종료가 아니라 atomic publication 이다

제안



foreground request 는 version 을 pin 하고 , deletion 은 같은 CAS word 를 전진시킨다

# 297-cell benchmark 는 lifecycle 선택을 비교한다

제안

# 297

logical cells



## 189

ROUTING × MEDIUM

6 deployable × 3 reuse × 3 pressure ×  
3 seeds = 162  
+ oracle on 27 operating cells

## 108

SCHEDULE × OPERATOR

2 schedules × 2 operators × 3 reuse  
× 3 pressure × 3 seeds

## 48

fixed blocks  
16 families ×  
3 seeds

## 131

global rows  
104  
continuous +  
27 RF

## 20k

blocked paired  
bootstrap

## ITT

failures score  
zero

UNEXECUTED · proposed, preregistration-ready · no PAPER-C result

# 리서치의 가치는 답보다 answerable question 을 만드는 데 있다

가설

NOW

- phase 정의
- prior mechanism
- capacity necessity

MATCHED EXPERIMENT

- destination routing
- cadence advantage
- NREM/REM factorization

LONG DEPLOYMENT

- lifetime equilibrium
- governance debt
- cross-hardware law

확장 가설

queue-before-storage · usable-capacity release · routing knee · clock-count crossover · recovery-substrate floor

# 좋은 sleep 연구는 실패했을 때도 무엇을 배웠는지 남긴다

제안

# 31

## adversarial findings

statistics  
capacity  
systems  
publication

NO DELAY BENEFIT

→ sleep-specific claim 철회

NO HYBRID DOMINANCE

→ matched frontier 만 보고

NO KNEE IDENTIFICATION

→ KNEE\_UNIDENTIFIED

NO HELD-OUT PREDICTION

→ curve, not law

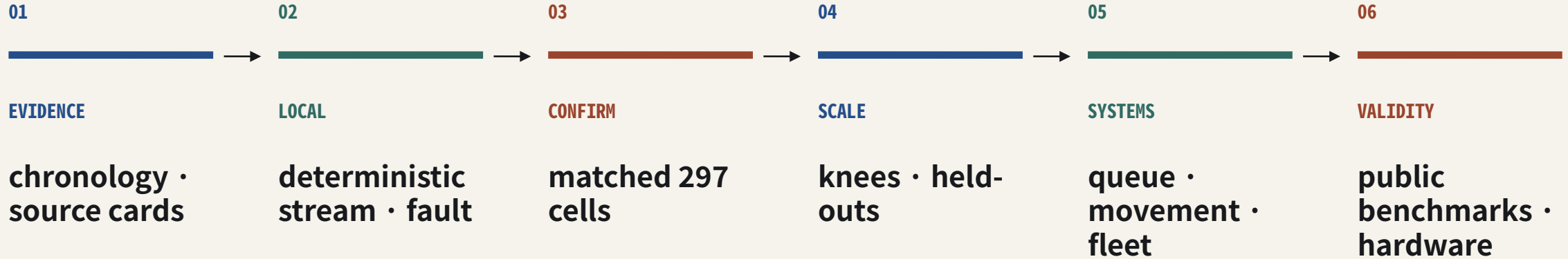
NO COMPLETE DELETE

→ stage 별 residual 보고

NULL ≠ FAILURE OF THE PROGRAM

# 다음 단계는 더 큰 모델이 아니라 더 단단한 lifecycle 실험이다

제안



**Minimum  
credible paper**

reproducible stream · matched destinations · predictive crossover or  
informative null · complete state · negative path

hardware run 과 human release gate 는 아직 수행되지 않았다

# Sleep 의 단위는 검증 가능한 기억 lifecycle 이다

통합 추론

CAPTURE

TRANSFORM

VERIFY

REUSE / UNDO



Wake 가 exact event 를 남기고  
Sleep 이 candidate 를 만들며  
Memory fabric 이 version 을 보유하고  
Control plane 이 publish 와 undo 를 결정한다

external vs parametric 은 철학이 아니라 lifetime routing decision 이다

# 핵심 1 차 문헌 · foundations & direct sleep

근거

- |    |                                                                     |    |                                                                |
|----|---------------------------------------------------------------------|----|----------------------------------------------------------------|
| 01 | Crick & Mitchison · Nature 1983 · dream reverse-learning hypothesis | 08 | Schwarz et al. · ICML 2018 · Progress & Compress               |
| 02 | Hinton et al. · Science 1995 · Wake-Sleep algorithm                 | 09 | Sun et al. · ICML 2020 · test-time training                    |
| 03 | McClelland et al. · 1995 · complementary learning systems           | 10 | Deperrois et al. · eLife 2022 · PAD                            |
| 04 | Robins · 1995 · pseudorehearsal                                     | 11 | Golden et al. · PLOS Computational Biology 2022 · SNN sleep    |
| 05 | Shin et al. · NeurIPS 2017 · deep generative replay                 | 12 | Tadros et al. · Nature Communications 2022 · sleep-like replay |
| 06 | Kamra et al. · 2017 · Deep Generative Dual Memory Network           | 13 | Harun et al. · TMLR 2023 · SIESTA                              |
| 07 | Kemker & Kanan · ICLR 2018 · FearNet                                | 14 | Packer et al. · 2023 · MemGPT                                  |

Links and exact evidence grades are retained in the companion paper/monograph.

# 핵심 1 차 문헌 · agent memory, capacity & systems

근거

|    |                                                          |    |                                                                         |
|----|----------------------------------------------------------|----|-------------------------------------------------------------------------|
| 15 | Lin et al. · 2025 · Sleep-time Compute                   | 22 | Dohare et al. · Nature 2024 · loss of plasticity                        |
| 16 | Rasmussen et al. · 2025 · Zep / temporal knowledge graph | 23 | Behrouz et al. · 2026 · Language Models Need Sleep                      |
| 17 | Mem0 · ECAI 2025 · memory operators                      | 24 | Kerestecioglu et al. · 2026 · Human-Inspired Memory Architecture        |
| 18 | Berges et al. · ICML 2025 · Memory Layers at Scale       | 25 | Auto-Dreamer · 2026 · learned external consolidation                    |
| 19 | Lin et al. · 2025 · Sparse Memory Finetuning             | 26 | RecMem / TiMem / GAM · ACL 2026 · agent-memory lifecycle                |
| 20 | Spens et al. · NeurIPS 2025 · learned offline processing | 27 | Useful Memories Become Faulty · 2026 · repeated rewrite counterevidence |
| 21 | Allen-Zhu & Li · 2024 · conditional knowledge capacity   | 28 | Agent Memory · 2026 · systems characterization                          |

BOUNDED AUDIT · non-discovery is not novelty evidence